AeroSentinel | Astro

- Deterministic satellite fault-recovery environment

- Five tasks; protected outcome scoring; separate verifier

- Demo task: eps_eclipse_recovery

What the agent must do

- Inspect symptoms and live telemetry

- Repair software controls while preserving hardware facts

- Recover power, temperature, pointing and downlink outcomes

State and actions

- Battery energy, temperature, scalar pointing, data and orbital time

- Read/write files, run bounded Python, advance orbit, finish

- Every reset reconstructs the seeded world

Simplified coupled model

- 95-minute orbit with sunlight, eclipse and a ground pass

- Energy balance and linear voltage; scalar pointing response

- Linearized heat loss and baud-limited communications

- Explicit approximation, not a flight-certification claim

Task depth and variation

- EPS, ADCS, thermal, RF software and two-fault cascade

- Seeded hardware values and mission energy targets

- Guided nominal settings plus protected mission constraints

Reward integrity

- Fixed success targets cannot be relaxed by agent settings

- Progress is normalized against the initial no-op outcome

- Catastrophic failures remain latched

- Final configurations are checked even after virtual time ends

Runtime isolation

- Verifier in harness; untrusted workers in sandbox

- Separate agent/evaluation UIDs; authenticated endpoints

- Read-only root, bounded storage, CPU, processes and output

- Detached workers killed before the next episode

Unknown-model resilience

- Same OpenAI-compatible endpoint contract for every model
- Malformed replies receive feedback and consume one step
- 40 replies, 10 calls per reply, 600-second execution cap
- Conservative context trimming preserves valid tool groups

Evidence and regression tests

- Present the generated gate.json and sanitized model logs

- Regression cases reproduce the scoring and reset defects

- Source fingerprints prevent stale evidence being relabeled

- A failed task is acceptable when correctly scored and explained

Live demonstration and limitations

- Run the demo; show telemetry, score components and termination

- Show the adversarial tests and a genuine failure trajectory

- Known limits: simplified physics and kernel/container reliance

- Submission: one Astro.zip with current-code evidence